

TITLE: A novel approach toward addressing keel bone concerns in extensive housing

INTRODUCTION: STATEMENT OF PROBLEM

Throughout the life of laying hens, 52-73% can undergo catastrophic and sometimes even fatal changes to their keel bone (KB)⁴, as shown in **Figure 1**. Keel bone damage (KBD) and keel bone fracture (KBF) are considered a major issue in the poultry industry affecting animal welfare as well as production. The transition from intensive (conventional cage) to extensive housing systems (enriched colony or cage-free) has been gaining momentum in the past ten years and will become more prominent in the next 25 years. Extensive housing systems provide more space and enrichments, e.g., perch, nest area, litter area, to allow hens to express natural behaviors.⁵ Future housing design will require gaining greater knowledge about the birds themselves and their living conditions. At present, there is insufficient knowledge of the root causes of KB developmental calamity.⁶ It is unclear if bone damage follows from a single large impact event during their life cycle, such as when navigating a 3D space, a low but constant force on the keel, for instance due to prolonged sitting, or due to a repeat loading as from daily egg production.⁷⁻¹⁰ How these events individually, or their potential interactions, shape animal welfare constitutes a major theme.

Currently there are limited ways to monitor the biomechanical conditions experienced by egg-laying hens. External impact and body-weight loads have been considered as a major cause. Recent measurements on biomechanical trauma conditions rely on acceleration ranges (g-values) or on sensing data from force platforms rather than on the more important direct force and impact pressure data measurements.^{9, 11-13} There is a need for the development of novel pressure sensing

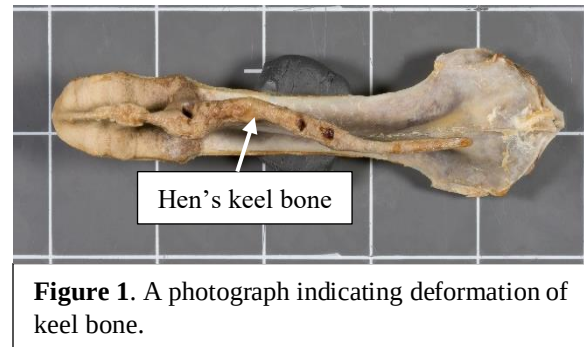


Figure 1. A photograph indicating deformation of keel bone.

systems for the use in egg-laying hens as commercial impact sensors provide less-than-ideal specifications for their applications in egg-laying hens. There are valuable examples of bone strain measurements. These have been performed in humans.¹⁴ Yet, such work has not been conducted in poultry in-vivo. Strain measurements in birds have been performed on muscle in-vivo¹⁵ and ex-vivo on poultry keel bone.¹⁶ The biomechanical conditions due to the egg-laying process itself present a larger challenge. Currently, no prior work exists to obtain keel bone stress and strain data associated with egg-laying, and to interrogate the role of egg-laying on KBD.

Predictive capabilities for biomechanical trauma conditions are state-of-the-art in the domain of human injury e.g., automotive accidents or in sports, through the use of Human Virtual Models. Thereby, physics-based simulations (finite element analysis to solve systems of partial differential equations) are employed to compute skeletal deformations and injury. This approach, despite its demonstrated capabilities, has not been applied to the study of KBD and KBF. Its realization would require input from assumed or measured acceleration, force or pressure data histories, accurate anatomical structure information as from 3D computer tomography (CT) imaging, and keel bone biomechanical properties. While such model components exist by themselves, the integration into a unified Virtual Bird Model (VBM) has not been performed.

We seek to transform the assessment of KBD and KBF by employing quantitative assessment methods enabled by leveraging sensor developments, computer simulation and tissue characterization, and to demonstrate the methodology to a critical example.

Our collaborative research team combines expertise in core poultry science (PI Karcher), sensor science (PI Nawrocki) and biomechanics (PI Siegmund). Together, our interdisciplinary team proposes to address the following overarching research question:

What are contributions and interactions of external and internal biomechanical loading on keel bone damage as experienced by an egg-laying hen. Specifically, we look to engineering technologies to ask:

- What are biomechanical loads on the keel bone as resulting from external loading?
- What are the stress and strain states in the keel bone as induced by egg-laying?
- How do external and internal loads interact to cause keel bone damage?
- How does stocking density affect keel bone conditions?

To answer this question, we will execute the following integrated specific aims:

SA1: Measure external loading on hens: We will employ 3D printed, custom force sensor technology to enable direct measurements of external force on birds from external loading (impact and body weight measured from skin mounted sensors).

SA2: Measure internal loading and strain on keel bones: We will extend our sensor technology to directly measure deformation conditions from internal loading (egg laying). We will implement direct bone mounted force and strain sensors for attachment to the keel bone, to measure external forces (impact) and bone deformation (egg laying).

SA3: The Virtual Bird Model: We will build the Virtual Bird Model (VBM) as a finite element model based on 3D imaging of bird anatomy and tissue biomechanical properties. We will employ VBM to compute the skeletal stress state under loading external and internal to the bird.

SA4: Measure and evaluate keel bone damage in extensive housing considering stocking density: We will integrate sensing and the Virtual Bird Model. We then apply the integrated approach in a setting to birds housed in an enriched colony setting and use stocking density as the housing design variable. We thereby will extend the sensing approach to monitor multiple birds over an extended time period and simultaneously sense forces at external and internal locations on birds together with bone strains. We will apply VBM to analyze the interaction between birds and of birds with enrichments to quantify the potential for keel bone damage.

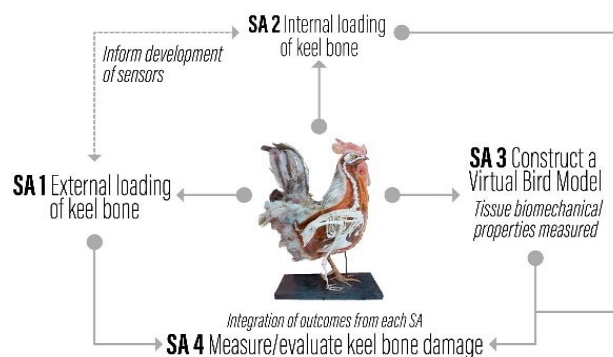


Figure 2. Specific Aims and their integration.

Overview of our Approach: Our interdisciplinary approach will apply multiple engineering technologies to experimental conditions in poultry studies, **Figure 2**. In the first two specific aims (SA1-2), we propose a unique combination of external (SA1) and internal (SA2) sensing modes to realize a condition monitoring approach in egg-laying hens. We will establish conformal sensors feeding data from external and internal loading into a computational bird model for computation of keel bone stresses. We will

conduct measurements on a white laying hen strain in two types of environments. In SA3, we establish the Virtual Bird Model (VBM) as a computational realization of the bird anatomy and endow the VBM with in-vivo measured biomechanical properties (using the Biodent instrumented reference point indentation probe of the keel bone and the in-vivo measured mechanical forces

(using conformal force sensors) experienced by hens. We validate the VBM through comparison of computed strain and measured strain data. Finally, in SA4 we then apply the condition monitoring approach to study a cohort of egg laying hens in an extensive housing system. The combination of investigator expertise in poultry science, sensor technology and computational biomechanics underpins this sustained interdisciplinary approach for translational impact.

Outcomes: We will acquire the ability to model keel bone quality and changes because of forces experienced during the laying hen life cycle in an enriched colony environment. Long term, the expected outcomes will provide knowledge to allow housing manufacturers to identify/develop materials that can be incorporated into housing structures to reduce the forces the keel experiences as the hen navigates 3D space (see letters of support from industry partners).

The proposed work aligns with the program area priority focusing on engineered devices, technologies and tools to understand keel bone damage. Our proposal is bold in that it intends to quantitatively understand the external (force from impact) and internal (stress from egg laying) forces applied to the laying hen keel bone. Our sensing approach and integration with the VBM is novel and no similar approach has been documented. Upon successful sensor deployment and VBM development, we will apply the integrated approach to analyze the history of forces exerted on the keel bone and interaction between multiple birds for keel bone damage.

RATIONALE & SIGNIFICANCE

Keel Bone Damage and Current Assessment. Keel bone deformities (KBD) were first reported in the 1930's and 1940's¹⁷⁻¹⁸ but disappeared from literature in the 1950's. In the early 1950's, the expansion of the egg industry resulted in a change in housing system from the deep litter floor to conventional cages. The skeletal problems of osteoporosis and cage-layer fatigue were first reported in 1955 as the conventional cage was becoming the “new” housing system.¹⁹ Reports of keel bone fractures and deformities were very limited attributing the issues to calcium, phosphorus and/or vitamin D deficiency. Roughly 95% of hens were in the conventional cage and 5% were in deep litter systems by 1980.²⁰ However, conventional cages eventually received greater criticism over the ethical concerns of hens kept in them²¹ and were banned in the European Union (EU) at the start of 2012²² making way for enriched or furnished cages and non-cage systems (single and multi-tier aviaries; free-range). Keel bone fractures and deformities have resurfaced in the recent years with the adoption of non-cage systems such as cage-free aviaries or free-range and barn systems. Currently of the 299 million laying hens, 35% are in cage-free housing but commitments from retailers and food service requires 72% of the U.S. laying hen population to be cage-free by January 2025.²³ The reported incidence of KBD varies around the world ranging from 50-97% (for recent review of literature⁶). Regardless of the exact percentage, one could anticipate 52-100 million US laying hens are impacted.

Technological advances over the past 10 years have changed how KBD is being assessed. Prior to 2015, keel palpation was the methodology used evaluating the bone for irregularities and callus formation.²⁴ Methods used today are computed tomography,²⁵⁻²⁸ radiography²⁹⁻³¹ and sonography.³⁰ A different approach to determine KBD explored the role of forces exerted on the keel.¹³ The main source of acceleration events (≥ 20 G-units) occurred when navigating perches or social interactions with conspecifics when collisions occurred.²⁶ The limitation was the use of tri-axial accelerometers positioned on the keel that recorded the kinetic events experienced (G-units). This novel approach summed all kinetic energy experienced at the keel but did not provide detailed

information on exactly where the forces were experienced. Each of these approaches have merit but a new approach is needed to better understand the etiology of KBD and potential for mitigation in future housing systems.

Sensors. Piezoelectricity³² enables electric polarization resulting from the application of mechanical stress: a piezoelectric material can generate electrical charges in response to external mechanical input. Poly(vinylidene difluoride) (PVdF)³³⁻³⁴ is an organic piezoelectric material that is widely used because of its low cost, ease of fabrication, physical flexibility and biocompatibility.³⁵ PVdF has been used to fabricate devices associated with sensing,³⁶ actuation,³⁷ energy harvesting³⁸ and energy storage³⁹.

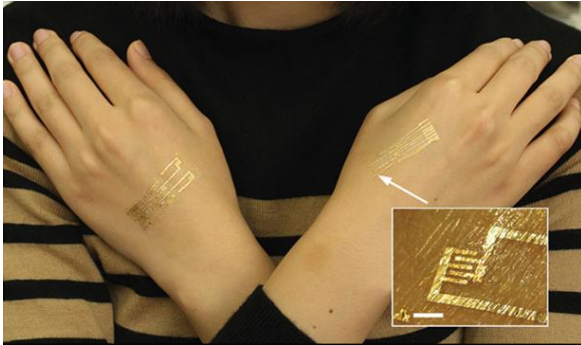


Figure 3. Conformal thin film electronics.

The polymer chain structures of the α - to β -phase and the crystalline and amorphous regions in poled PVdF are illustrated in **Figure 4a**. Normally, the non-polar thermally stable α -phase is the dominant phase in initially manufactured PVdF films. The phase transformation from the α - to β -phase, which is of primary interest for piezoelectric property, occurs through stretching⁴⁰ or polarization.⁴¹ Due to the random orientation of the dipoles in the α -phase, the dipoles cancel each other out resulting in neutral surface charges. The crystalline structure of β -phase can be obtained by mechanical stretching along one direction and poling by applying a strong electric field at elevated temperatures, then lowering the temperature to lock the domains in their poled state. During the stretching process, the randomly oriented crystalline α -phase changes into a semi-crystalline β -phase due to a strong mechanical orientation effect. The polarization aims at rotating the molecular dipoles towards a common direction.

The piezoelectric charge outputs from the sensing device upon application and release of external forces is converted into voltage through a charge amplifier. This extracts the charges generated by the PVdF sensing device and converts these to a high impedance signal from a piezoelectric sensor to low impedance voltage signals (**Figure 4b**).^{36, 42} The feedback capacitor in the charge amplifier circuit produces an output voltage proportional to the integral of current with respect to time (i.e., charge generated by a piezoelectric PVdF sensing

Piezoelectric PVdF offer many advantages when constructing sensors to detect pressure or force. Thin films can be made ultra-thin (few microns), allowing for exceedingly conformal contact with highly irregular surfaces, such as skin (see **Figure 3**) or bones. In addition, they can be made to continuously cover a large area. Finally, such a film can be patterned into a matrix of individual sensors allowing for detection of a (spatial) distribution of forces.

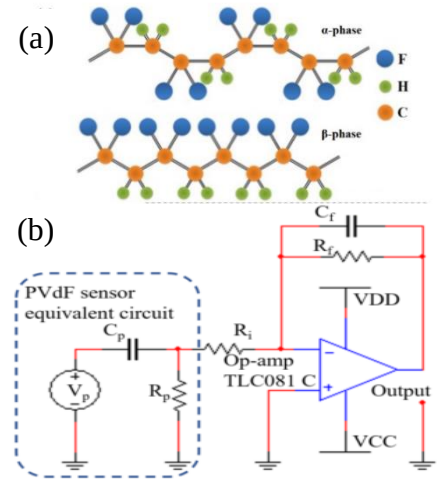


Figure 4. (a) α - and β -phase structures of PVdF polymer. (b) Charge amplifying circuit used to convert the charges extracted by the PVdF film to a voltage output.²

device). The expression for the output voltage of the charge amplifier circuit is: $V_{out} = -\frac{1}{C_f} \int I(t)dt = -\frac{Q}{C_f}$, where C_f is the capacitance of feedback capacitor.⁴²⁻⁴³ The resistor in the feedback path prevents the op amp from saturation.

3D Printed PVdF Force Sensors: Preliminary Data. We have developed a new sensor fabrication technique that allows to make highly sensitive sensors of any shape (including folds around sharp edges).^{2, 43} Fully functioning sensors are manufactured by a combination of 3D printing techniques, namely Fused Deposition Modelling (FDM) (to fabricate PVdF), and Direct Ink Writing (DIW) (for precision deposition of silver ink acting as electrodes used to extract the charges). An additional advantage of using DIW is to define the geometry of individual sensors (so called “direct patterning”), to allow for determining the distribution of force across an area. We have pioneered a method of improving the ratio of β - to α -phase, critical in improving sensor sensitivity, that is based on application of electric field to PVdF film during sensor fabrication. FDM of PVdF involves extrusion of heat-assisted liquified PVdF filament wire. When PVdF is in a liquid form, the film expands loosening entanglement of individual PVdF polymeric chains, allowing them a freedom of movement and re-orientation. PVdF molecules possessing an electric dipole are susceptible to external electric field and attempt to align their orientation with the electric field, analogous to operation principle of a compass. PVdF is extruded to form a thin film, which results in quick cooling of the polymer, still in presence of the electric field, which results in re-oriented polymers to form favorable β -phase alignment and improve sensor sensitivity.

Figure 5 demonstrates results of electromechanical characterization of our 3D printed PVdF film. **Figure 5a** shows series of recorded voltages due to the application (positive spikes) and removal (negative spikes) of external force. Application of larger force produces a higher spike. **Figure 5b** shows a calibration curve of PVdF sensors. As a piezoelectric material, PVdF cannot be used to detect static force; it only responds to dynamic or change in force. As such, for instance, it cannot be used to detect forces when a hen is sitting. However, such information can be determined from the fact that a removal of force produces a negative voltage spike (**Figure 5b**).

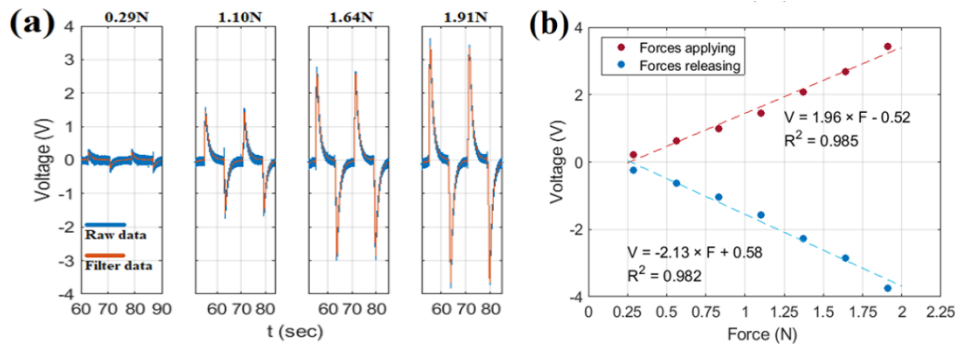


Figure 5. (a) Piezoelectric output voltages of a 3D printed PVdF sensing device in response to applying / releasing forces, (b) the calibration curve of the 3D printed PVdF sensing device.²

Three types of sheet-like PVdF sensors have been demonstrated: thick-film ($>100 \mu\text{m}$), thin-film ($<100 \mu\text{m}$), and nanomesh, each with their own advantages and disadvantages. Thick- and thin-film sensors are already commercially available. Advantage of thin-film sensors is that they offer higher sensitivity and conformability to irregular 3D shapes, such as animal skin or bones. However, thinner films are also more fragile and are susceptible to being damaged by the bird, for instance due to pecking or physical contact with the environment (e.g. when repeatedly sitting and moving around). As such, they might be best used for short-term, on-skin data collection. Thicker

sensors, while less sensitive and less conformal, are also much more resilient against physical damage. As previously mentioned, the group of the PI Nawrocki, has developed a 3D printing process to fabricate such sensors⁴³ (**Figure 6**). A main advantage of such a fabrication approach is that it allows for an arbitrary shape. We believe that this will make it easier to attach the sensor to the highly irregular keel bone to more accurately measure the forces impacted on it, for instance by minimizing the effects of sensor bending or folding; we have verified that bending does not negatively affect sensor performance.¹

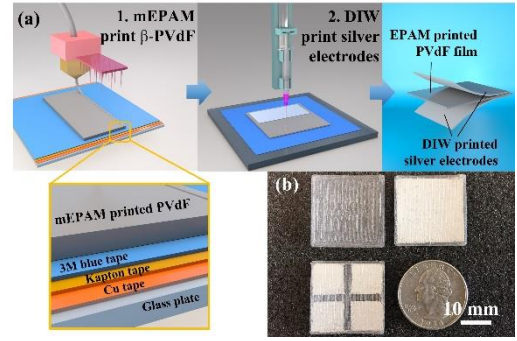


Figure 6. (a) A single 3D printed PVdF film, (b) 3D printed PVdF films with different silver electrode patterns.¹

Conformal *strain* sensors will be fabricated using a similar thin film technology, with a standard strain gauge design measuring change to electrical resistance. Our custom-made thin-film strain gauge sensors will be made to be a few hundred nanometers thin, making them imperceptible to the bird, but not interfering with keel bone-mounted PVdF force sensors.

Nanomesh PVdF Force Sensor: Preliminary Data.

Nanomesh (**Figure 7**) is a structure consisting of threads or fibers with diameters in the nanometer range.⁴⁴ A third type of PVdF sensor is based on nanomesh structure.⁴⁵ In contrast to thin films, nanomesh structures offer two main advantages: stretchability and gas/liquid permeability. Nanomesh structure consists of intersecting threads or fibers with diameters in the nanometer range. The individual threads are attached to other threads only at the point of their intersections, allowing for the structure to be easily stretched (in excess of one hundred percent⁴⁶) without affecting sensors' mechanical or electrical performance. The empty space between the threads allows for gas or liquid to easily pass through. While the most fragile of all the sensors, stretchability and liquid permeability would be advantageous for long-term *in-vivo* data collection.

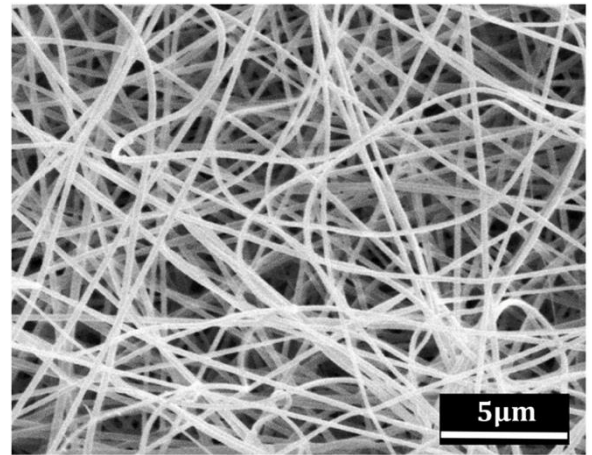


Figure 7. Nanomesh fabricated in our laboratory.

It has found a number of applications, including optical sensors,⁴⁷ pressure sensors,⁴⁸ air filters,⁴⁹ oil-water separation,⁵⁰ rechargeable battery electrodes,⁵¹ tissue engineering,⁵² and on-skin biosensors.⁵³ Nanomesh fabrication typically requires polymers, long molecules connected via weak covalent bonds, for fabrication of long nanomesh threads. Both natural (silk fibroin and gelatin) and synthetic polymers (polylactic acid, polyurethane, polymethyl methacrylate, polyvinyl alcohol (PVA)) have been shown to produce nanomesh threads. A number of advantages of nanomesh, as compared with solid thin films, include large surface area-to-volume ratio, high

porosity, significant mechanical strength including stretchability (in excess of 150%⁴⁶), as well as high elasticity (Young's modulus of ~ 1 -100 MPa⁵⁴⁻⁵⁶) and gas or liquid permeability.⁵³ The size of a nanomesh thread, including its length and diameter, typically depends on the polymer used and the fabrication method. While a number of fabrication techniques exist, including thermally-induced phase separation, template synthesis, and self-assembly, electrospinning is used most often due to its relatively simple setup, ability to mass-produce continuous threads, and ability to control thread diameter, composition, and orientation. The electrospinning setup consists of a high voltage supply, a capillary tube (usually a syringe for holding polymer solution) terminated with a metal needle, and a collecting plate or drum. When a high electric field is applied between the needle and the collecting plate, a charge is applied on the surface of the polymeric solution, which is initially held by surface tension. As the electric field increases, the hemispherical surface elongates forming a conical shape known as a Taylor cone.⁵⁷ When the repulsive electrostatic force overcomes the surface tension, the charged jet of fluid is ejected towards the collecting plate. As the ejected polymer solution is highly unstable, it elongates to form extremely long and thin threads. Due to the high surface area-to-volume ratio, the solvent quickly evaporates, and the polymer threads solidify in a randomly oriented fashion on the collecting medium.

As previously discussed, a sandwich piezoelectric force sensor consists of the following elements (**Figure 6a**): (1) *piezoelectric material* (in our case piezoelectric polymer PVdF) acting as the active medium; and (2) *electrodes*, typically made from metals, such as Au. As shown in **Figure 8a**, a nanomesh piezoelectric force sensor will contain the same elements, albeit made from nanomesh instead of a solid film. We will first fabricate large-scale sensors with features (size of individual sensors) down to a few hundred microns. As discussed in the previous section, only electrospinning allows fabricating long and continuous nanomesh threads and is the most suitable approach to fabricate from piezoelectric polymers (*thermal-induce phase separation* only works with natural polymers, and *self-assembly* and *template synthesis* results in short threads⁵⁸⁻⁶¹). As PVdF is a poor electrical conductor, direct electrospinning fabrication of PVdF nanofibers is possible: electrospinning requires application of ultra-high voltages, in excess of 50 kV, which is advantageous for PVdF as it aids in α - to β -phase transition, improving its sensitivity (piezoelectric constant). The following describes the proposed fabrication steps: (1) We will electrospin an PVdF nanofibers. Next, (2) we will obtain both, top and bottom, electrodes by directly evaporating metal (Au) onto and bottom, electrodes by directly evaporating metal (Au) onto PVdF nanofibers (**Figure 8a**). A matrix of individual sensors, called “pixels”, necessary to obtain a distribution of forces acting on keel bone, will be achieved with the use of a shadow mask to define their geometries, specifically geometries of individual electrodes (to allow measuring spatial distribution of force), as shown in

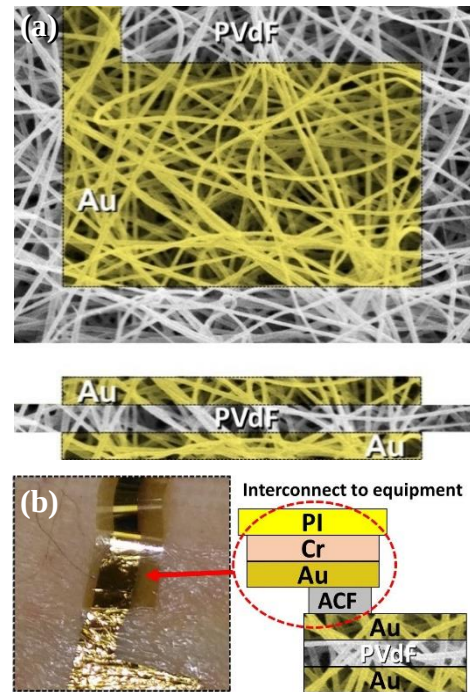


Figure 8. (a) Nanomesh PVdF force sensor with gold electrodes: top is top view, and bottom is cross section. (b) Flexible interconnect used to connect sensor with external equipment.³

Figure 6b. The structure will be encapsulated with Parylene (commercially available, biocompatible and FDA approved encapsulant⁶²⁻⁶⁴) to avoid cross talk between individual sensors (not shown in the figure).

To simplify the fabrication process, instead of evaporating metal electrodes on PVdF nanofibers, (top and bottom) metal electrode arrays would be fabricated directly onto Parylene layers and then sandwiched together. While simpler to fabricate and more durable, this option would produce non-stretchable and non-permeable sensors. Finally, the data from the sensors will be read using flexible interconnects that will be used to bridge the electrical gap between fragile nanomesh sensors and electrical wires connected directly to the electronic equipment, such as parameter analyzer or RLC meter. The interconnects will be made from thin film metal evaporated onto flexible, 50 μm thin polyimide tape, as we previously demonstrated with thin-film biopotential sensors.³ The sensors will be connected with the interconnects using anisotropic conductive film (ACF) to simplify the connection process (**Figure 8b**).

Sensor Mounting on Live Birds: Preliminary Data. We have performed preliminary work into the feasibility of the use of our 3D printed PVdF force sensor and our nanomesh PVdF force sensors to collect external loading data (**Figure 9**). Dummy sensors (PVdF polymer, without electrodes; future sensors would have to be encapsulated with a biocompatible polymer, such as Parylene, in order to avoid direct contact of skin with metal electrodes) were externally attached to a bird's skin, around keel bone area, using a double-sided tape followed by additional application of superglue. The sensors stayed in place for up to 6 hours without negatively affecting birds' behavior. We have discovered that in order to obtain a conformal contact around keel bone, the 3D printed sensors had to be folded, resulting in a visible crease. However, in a laboratory environment we have verified that a proper calibration can effectively account for such a physical deformation.¹ Also, our next generation of printed sensors will include bone curvature in their design, to avoid the folding issue. Furthermore, it can be seen that nanomesh sensors were able to conformally wrap around the curvature of the bone, with most of the thickness being the medical glue.

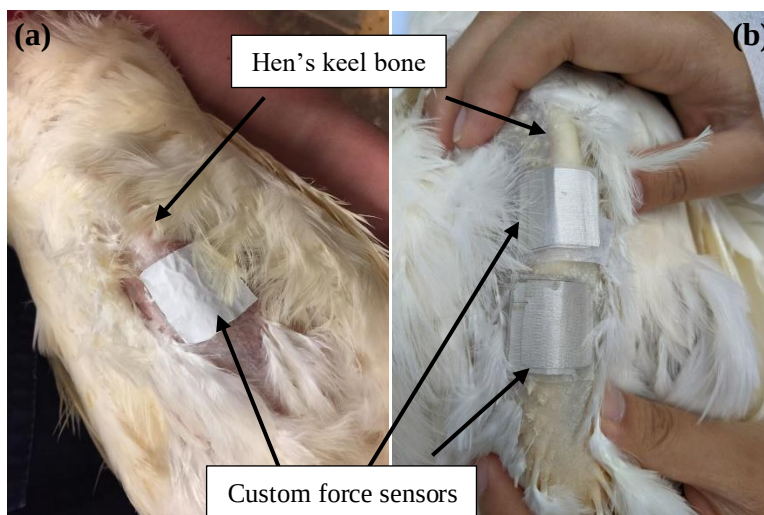


Figure 9. Externally mounted (a) nanomesh and (b) 3D printed force sensors, placed around keel bone area of a live hen.

Virtual Models: Preliminary data. In silico investigations of injuries in impact and crash scenarios are viable tools to assess the safety of engineered systems where an interaction of a living subject with a physical environment occurs.⁶⁵ In fact, the modern design of automotive transportation systems with regards to passenger safety and pedestrian safety could hardly be envisioned to be conducted without such simulations. Today's simulations codes for the human body and the crash body (car, barrier) are advanced to a very high fidelity and level of validity.⁶⁶

The human body, in particular, required to be defined by both a hard tissue (bone) skeleton embedded into the soft tissue surrounding.⁶⁷ In this context we know that the use of advanced finite element models allows for accurate prediction of injury.⁶⁸ To realize such insights key aspects are necessary: impact forces and acceleration data, anatomical 3D geometries, as well as biomechanical properties of both soft and hard tissue. In human crash investigations, experiments with engineered crash test dummies plays a central role.⁶⁹ Such devices are used to measure forces and accelerations but are also used in the validation of models. We know that the use of engineered crash test dummies has limitations. It is understood that accurate injury predictions require a series of such dummies, accounting for size and weight changes. Accurate models, however, do need to go beyond the engineered dummies. It is recognized that the actual interaction of the living subject with the environment can affect safety. Consequently, non-critical simulated accident conditions with living or post-mortem subjects have been realized.⁷⁰⁻⁷²

In the computational analysis of injury, the realization of the anatomical features plays an important role. We know that the most accurate model representation of individual subjects can be realized using image-based models.⁷³ The pathway to such models has been demonstrated as containing a 3D volumetric image acquisition (CT scans), the translation of such a geometry to a finite element mesh, the definition of actual or estimated material properties to hard and soft tissue, a hard tissue failure criterion, as well as the definition of initial and boundary conditions. Currently, available finite element solvers can compute the response of complex 3D finite element models under nonlinear conditions and 10 million degrees of freedom in day-to-day computing-systems operations.

Underpinning the accurate prediction of bone failure is an increased understanding of the biomechanical properties of bone together with a realization of a deformation state and of a failure of hard tissue. Recent research findings have demonstrated that the in-vivo measurement of bone biomechanical properties can be accomplished by the use of the reference-point indentation (RPI) method.⁷⁴ This method provides rich force-indentation depth data. As with any indentation method for the determination of mechanical properties, it is necessary to employ indentation analysis models to translate the measured raw data of force-indentation depth into meaningful stress-strain data. In particular, for problems concerning bone injury and fracture, it is relevant to discern the tissue failure properties from indentation data. To that end one can apply either stress invariant criteria,⁷⁵ damage mechanics,⁷⁶ or fracture mechanics.⁷⁷

We have demonstrated a process to generate simulation models for the interrogation of keel bone fracture under either external (impact load) or internal (egg-laying) conditions. We obtain CT-scans of egg-laying hens without visible fracture. CT scan data is reconstructed and converted into a finite element mesh (using the code Synopsys' SimplewareTM software⁷⁸). The keel is segmented into bone and cartilage components. A finite element discretization is constructed, preferentially using 10-node tetrahedra elements. A typical model of this kind employs of the order of 25,000 to 50,000 finite elements in a keel bone model. The keel

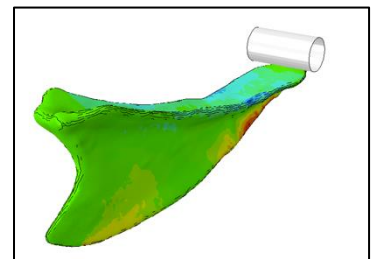
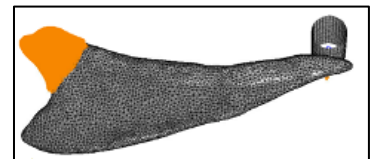


Figure 10: Example of stress analysis of keel bone as obtained by the finite element analysis. Red contours indicate high levels of stress and propensity to bone damage.

bone is endowed with estimated bone properties obtained by translating gray-scale image information into property data following standard processes of bone biomechanics. Bone and cartilage possess distinctly different elastic properties, with cartilage possessing an elastic modulus about one order of magnitude of that of bone. Simulations of the biomechanical stress-strain conditions in the keel bone we employ a high-end commercial finite element code (Simulia's ABAQUS software⁷⁹). With this code, we conduct a stress-strain analyzes of the keel bone and compare keel bone stresses to damage criteria. For example, one can consider the criterion of the maximum principal stress for the onset of fracture formation. **Figure 10** depict the key components of a typical analysis.⁸⁰ A model of a keel bone is shown as discretized by tetrahedral finite elements. The displacement boundary conditions will mimic anatomical conditions. The keel is considered as fixed at its fulcrum-near termination and is free at the opposite (cartilage) end. In order to mimic the path an egg would take, and to introduce associated loads applied to the keel bone, we define a rigid body. We translate this rigid body along the egg-laying path, thereby seeking to approximate the force exerted onto the keel bone. For impact load on keel bones we will employ sensor-measured force data as the load input to the simulation models.

Reference Point Indentation.

The in-vivo RPI technique to obtain the biomechanical properties of the keel bone will be used. This approach employs the commercially available BioDent⁸¹ system.^{74, 82} In the RPI technique a hollow needle is inserted through skin and muscle to the surface of the bone and provides a reference position.

Then, a needle inside the hollow needle is used to interrogate the bone tissue in a force and depth sensing

measurement. Using established protocols based on the mechanics of indentation,⁸³ we translate RPI data (force-indentation depth) into bone elastic moduli, and into estimates of bone failure properties (strain-to-failure).⁸⁴ As an example of the initial use of the RPI method in poultry, **Table I** above, capture the indentation distance increase (IDI) and total indentation distance (TID) results from RPI on conventionally-caged laying hen bones (N = 153 total bones). Statistical differences were found when evaluating bone properties compared between a lighting treatment (control vs PAWS), bone type (femur, humerus, and tibia), and location of the RPI measurement along the bone (distal, midpoint, or proximal). No differences were observed in bone properties between age (32 vs 52 weeks of age) in either IDI or TID.

Table I

		IDI (mean±SE)	P-value	TID (mean±SE)	P-value
Treatment	Control	6.73±0.11 ^a	0.09	58.30±0.61	
	PAWS	6.48±0.10 ^b		57.41±0.54	
Bone	Femur	6.60±0.12		58.84±0.68 ^a	≤0.05
	Humerus	6.78±0.13		58.35±0.76 ^a	
	Tibia	6.43±0.12		56.39±0.67 ^b	
Location	Distal	6.30±0.12 ^a	<0.0001	6.30±0.12 ^a	<0.0001
	Midpoint	6.35±0.12 ^a		6.35±0.12 ^a	
	Proximal	7.17±0.12 ^b		7.17±0.12 ^b	

APPROACH

ANIMAL HUSBANDRY

White feathered laying hens will be raised in conventional cages (SA1-2) and enriched colony cages (SA4) at the laying hen facility (Purdue University). Hens will be housed at a minimum of

432 cm²/hen with food and water *ad libitum*. Laying hens will be individually housed for SA1 and SA2 for 1 to 2 days initially. As the sensors are developed and tested, the laying hens will be housed for a longer period of monitoring (up to 14 days). For SA4, the laying hens will be housed in enriched colony cages, which will provide perches, nesting area and larger cage size in width, length and height. This housing system can accommodate 20 birds as a group size. Groups of 5, 10, 15, and 20 hens will be housed for 7 days repeating this approach during SA4. As the trial is repeated longer periods of monitoring may occur (up to 21 days). All experimental procedures will be approved by the Purdue Institutional Animal Care and Use Committee (PACUC) prior to the start of any work.

SA1: MEASURE EXTERNAL LOADING ON HENS

SA1 Outcomes: Direct measurement of impact pressure on keel bones through conformal pressure sensors will provide quantitative answers on impact forces experienced by birds due to interaction with external objects.

SA1a – External Force Sensor Validation (PIs Nawrocki, Karcher): The goal of this aim is to ensure that we can successfully *collect force data from externally placed sensors*, on birds' skin. Here we look to collect data to determine if keel bone fractures are due to sporadically experienced large force, such as during animal collision with other birds or stationary objects. Our piezoelectric PVdF have been demonstrated to successfully translate forces onto measurable quantities, such as voltage. However, these tests were performed in highly controllable, “sterile” laboratory environment. Performing such measurements on soft biological bodies adds complexities, noise, and other difficult to control factors. Performing such measurements on live birds adds further challenges: we need to demonstrate that these sensors can: (i) be successfully mounted on birds (our preliminary analysis (**Figure 9**) only showed “dummy” PVdF films and not entire sensors), (ii) without adverse effects, (iii) successfully collect force data, and (iv) survive (not being destroyed by birds' behavior) for a given amount of time. Additionally, sensory data will have to be recorded, stored, and retrieved for later analysis without adversely affecting birds' behavior.

Experimental Design 1.1: We will fabricate, using our previously developed 3D printing technique, custom-made PVdF force sensors that will be fit onto bird's skin directly atop of the keel bone. The printed sheet, fit with an array of sensors, will be approximately 3 cm x 12 cm, to cover the entire keel bone area. This sheet will be attached to the bird's skin using a double-sided medical adhesive tape or surgical glue (**Figure 9**). The force data from the sensors will be collected using a commercial development board. This board will be fit with customizable electronics, to convert charges generated by PVdF film into voltage (circuit is shown in **Figure 4d**), as well as a microprocessor to store the data locally, and a wireless transmitter for external data analysis. The board, approximately 5 cm x 7 cm, weighting 10 grams, is small and light enough to be fit into a portable “backpack” that will be mounted onto the bird's back, between their wings, as reported in previous works.^{13, 85} The PVdF sensor sheet will be connected to the board using thin film interconnect wires running along the side of the bird (**Figure 8b**).

Experimental Design 1.2. To validate the recorded data from the previous experiment, an array of PVdF force sensors will be placed externally, around the keel bone area, with a known force being applied to the sensors. Data will be recorded to make sure that correct force can be measured.

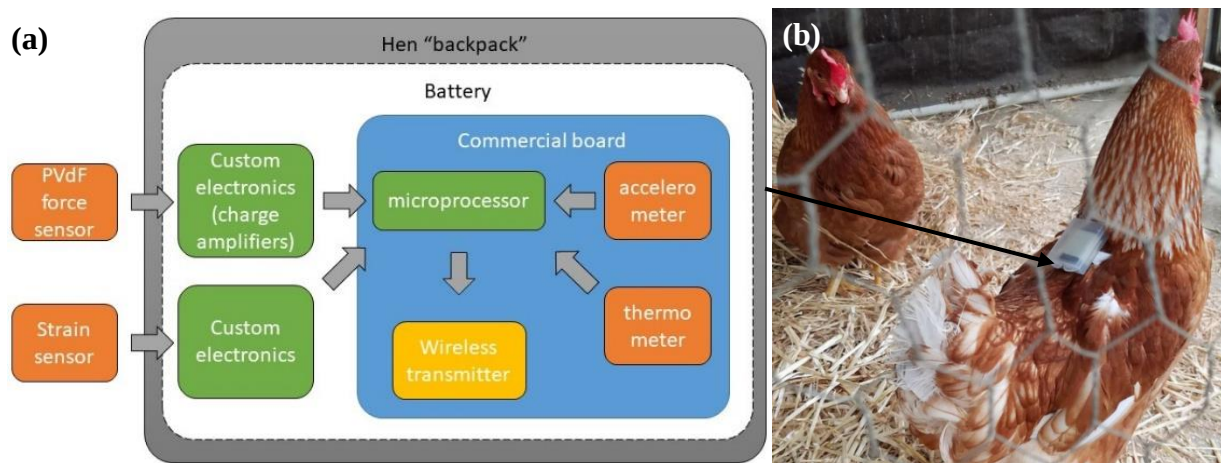


Figure 11. (a) Force sensory data will be collected, stored, and transmitted using a commercial development board, that (b) will be placed in a small “backpack” placed between bird’s wings.

SA1b - Data Collection and Retrieval (PIs Nawrocki, Karcher): Force sensory data (voltage), will be pre-processed (via charge amplifier and a filter, **Figure 4b**) and stored locally, before being downloaded for further processing and analysis, see **Figure 11a**. For this purpose we will utilize a commercial board, such as CC3200-LAUNCHXL from Texas Instruments,⁷⁹ or The Bean from PunchThrough,⁸⁶ fit with custom electronics (charge amplifier), memory (data storage), and wireless transmitter (data retrieval). The board will be fit with additional commercial sensors, namely accelerometer and thermometer. With all of the data being timestamped, this will enable gaining additional insights, for example, correlation of measured forces with bird movements, such as when a caretaker enters the room. To validate the timestamped data, the room will be mounted with cameras to capture the bird from multiple angles. **Similar to prior studies, behavioral data and identification of force events will be verified with the video recording.**¹³ The board will be housed in a previously developed and used “backpack”, as previously described.

Pitfalls and Limitations. Our laboratory testing shows that 3D printed thin-film PVdF films can accurately correlate magnitude of applied force to the magnitude of recorded voltage. **However, we do not yet know if our sensors are sensitive enough to detect impact forces through the skin mounted sensor. As a backup strategy, we would use our high-sensitivity nanomesh PVdF sensors in a skin-mounted configuration.** Our preliminary investigations indicate that 3D printed thin-film PVdF sensors can remain attach to skin for at least 6 hours. We will carefully monitor for sensor delamination from the skin to ensure viable sensor attachment over the needed time span to acquire meaningful impact data. If sensor delamination occurs, we will periodically replace the sensors.

Deliverables.

- Validation of PVdF piezoelectric thin film sensors to collect force data from live animals
- Validation of data recorded from external sensor to detect external loading

SA2: MEASURE AND EVALUATE LOADING AND STRAIN ON KEEL BONES

SA2 Outcomes: Direct measurement of strains on the keel bone will provide quantitative answers on biomechanical forces experienced by a bird load during egg-laying. We will be able to assess strain states for the keel bone due to internal loads.

SA2a – Strain Sensor Calibration (PIs Nawrocki, Karcher, Siegmund): To complement our knowledge from SA1 on forces applied to the keel bone, here we aim to collect data on actual bone strains. This will enable us to measure conditions under both externally applied, large but sporadically occurring forces (collisions), and also due to internal loads, commonly due to long-term application of small repeated forces, such as due to the bird's weight or laying eggs. For this purpose, we aim to apply a set of two **strain** sensors, placed along the X and Y of the keel bone. The conformal sensors will be surgically attached directly onto the bone, to accurately measure bone strains on the keel surface. Strain sensor data will be calibrated by attaching strain sensors to (i) 3D printed models of keel bones, and (2) on excised keel bones, and with known force applied to the (model) bone.

Experimental Design 2.1: Keel bone will be (i) created as 3D printed models or (ii) extracted from a euthanized bird. In each case, two strain sensors will be attached to the bone surface, to measure the strain along both of the surface tangential directions. A known mechanical force will be applied to the bone, and well-defined displacement boundary conditions will be considered as supports. Strain data will be collected from the sensors. We execute finite element models (SA3) of the keel bone models considering either model material data for the 3D printed keel model or best estimate bone tissue material data for the excised keel. The finite element analysis then provides theoretical, best estimate strain data. The comparison of measured and computed strain data is then used to calibrate the sensor output.

SA2b – In-vivo Strain Sensor (PIs Nawrocki, Karcher): To collect the actual strain experienced by the keel bone due to loading under in-vivo conditions, a set of strain sensors will be surgically placed on the keel bone. An individual with years of surgical experience with poultry will be assisting with sensor placement. Additionally, the PACUC attending veterinarian will be overseeing and assisting during surgical procedures (see letters of collaboration). Measured data will be correlated with previously collected calibration data from SA2a to obtain strain data in-vivo.

Analogously to PVdF force sensors (**SA1a/b**), the data will be collected, stored, and transmitted using commercial board placed on birds' backs. The communication between the strain sensors and the board will be via aforementioned thin film interconnects. To minimize bird discomfort and make the surgery simpler, the thin film interconnects will access the sub-dermally placed sensors at the place of skin incision and will be connected externally with the "backpack".

Experimental Design 2.2: Experiment will include surgically inserting two separate strain sensors, which will be attached to the bone using surgical glue.⁸⁷⁻⁸⁸ Initially, commercial thicker film strain sensors will be used to expedite the data collection. Thinner, custom-designed to fit the irregular shape of the keel bone sensor set, capable of simultaneously measuring X and Y strain, will be fabricated in-house. They will be more flexible and more comfortable/compliant, to improve the quality of data and animal comfort.

SA2c – Force on Keel Bone and Strain Sensor (PIs Nawrocki, Karcher): Our general aim is to most accurately collect the data impacted on the keel bone. One of the main advantages of the previously described 3D PVdF force sensor is in the ease of use, namely external application on bird's skin. However, force impacted on the skin is not the actual force impacted on the bone. It is not known how long such a sensor could be deployed, which would make it difficult to correlate the force data with observed developmental changes. Additionally, our 3D printed sensors are

somewhat thick. While this is advantageous in making them last longer in bird housing environments, it will dampen the force impacted on the bone and certainly affect the changes to the bone. Therefore, an imperceptible, long-term monitoring solution is required.

We propose to develop a new type of force sensor with the aim of deploying it directly onto the surface of the keel bone. The nanomesh PVdF sensor will be made from piezoelectric PVdF using previously discussed electrospinning technique. Such fragile yet imperceptible sensors will be surgically laminated on the bone allowing for better accuracy of force measurement. In combination with ultra-thin film strain sensor, the stretchable and biocompatible nature of nanomesh sensor should allow for long term (up to 1 month) monitoring of both, impact and egg laying, forces on the keel bone. The development of nanomesh PVdF sensor will be expedited by our previous thin-film (**Figure 6**) and nanomesh (**Figure 8**) PVdF sensor efforts.

Experimental Design 2.3: Stretchable, and permeable force sensors will be made from piezoelectric PVdF using electrospinning technique that produces exceedingly long (up to few meters) but thin (down to hundreds of nanometers) fibers. This produces a tissue-like structure that can gently wrap around arbitrary 3D shaped structure, such as a keel bone. We will fabricate conformal thin film and nanomesh force sensors as well as conformal thin film strain sensors. Both, force and strain sensors will be surgically placed underneath the skin, directly laminated on the keel bone. Several small incisions will be made on the bird's skin to get access to the bone. The PACUC attending veterinarian will be consulted for minimizing the location and size of incisions made to allow for sensor attachment. The sensors will be attached to the opposite corners of the bone using surgical glue.⁸⁹⁻⁹⁰ Previously described thin film interconnects will be used to connect the implanted sensors to the shoulder placed backpack housing the data storage board.

Pitfalls and Limitations. The sensor placement directly onto the keel bone adds complexity. If surgical attachment emerges as to ensure appropriate surgical procedures, we can expand our pre-surgical practice, both on work to attached sensor to excised keel bones, as well as the number of sacrificed birds for surgery and sensor attachment training. The in-bone implanted sensors may result in alteration of the bird's behavior after the surgery. However, the creation of a custom-designed sensor should alleviate that possibility.

Deliverables (for SA2).

- Validation of strain sensor data from excised and internal keel bone
- Validation of force sensing directly on keel bone

SA3: VIRTUAL BIRD MODEL

SA3 Outcomes: We will establish a virtual model of a bird consisting of main hard and soft tissue structures. The model geometry is derived from 3D imaging of birds. Biomechanical properties of keel bones are determined by in-vivo measurements. The model predicts stresses and strains in the keel bone from external and internal loading.

SA3a: 3D Imaging and Anatomical Reconstruction (PIs Karcher, Siegmund): We will conduct full 3D computed tomography (CT) of 6 birds from the study as housed in conventional cage. Imaging will be conducted following the protocol developed in the Karcher lab.^{27, 91-93} Following imaging, 3D image reconstruction (using the commercial software package Simpleware) will be used to obtain the 3D skeletal structure and the pectoralis muscle. Image segmentation of the keel

bone alone by absorption related image greyscales allows us to assign cartilage and bone domains to the keel bone. Figure 12 depicts a preliminary example for the type of anatomical reconstruction we seek to obtain. The combination of soft and hard tissue constitutes the key geometric domain of the VBM.

SA3b: Keel Bone Biomechanical Properties (PIs Karcher, Siegmund): The effort of biomechanical characterization will focus on the keel bone. The in-vivo reference indentation method (RPI) technique employs the commercially available BioDent⁸¹ system.^{74, 82} To obtain information on the spatial gradation in keel bone biomechanical properties we will conduct RPI at multiple sites along the keel bone. We will validate the approach on reference samples extracted from the keel bone on which we will first perform RPI measurements followed by standard 3-point bend experiments on beam shaped specimens extracted from the keel bone on a mechanical load frame. For the characterization of the pectoralis muscle, we will use excised muscle tissue as current in-vivo characterization methods for soft tissue to provide limited information on the depth dependence of properties. The hyperelastic passive tissue properties will be measured in biaxial tensile tests employing a bi-axial tensile load system specifically constructed for soft tissue characterization.⁹⁴ Measured force-tissue elongation data will be translated to the elastic constitutive parameters using standard protocols.

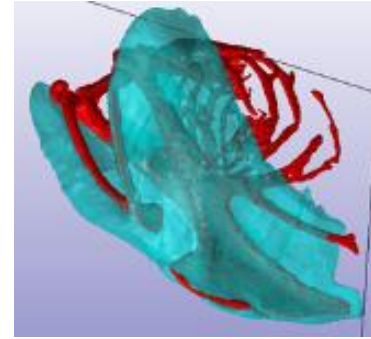


Figure 12: A tentative virtual bird model consisting of main skeletal bones and the breast muscle.

SA3c: Virtual Bird Model (VBM) (PI Siegmund): We will establish the Virtual Bird Model as a finite element model. The model geometry is obtained from the reconstructed animal anatomy, which is discretized by a finite element mesh using a commercial software package (Simpleware). This approach will use an unstructured mesh with tetrahedra elements of quadratic shape functions. The model is endowed by tissue biomechanical property data as per SA3b. We then solve for deformation, strains, and stresses based on assumed biomechanical boundary conditions under quasi-static loading conditions. This is like what would occur during a palpation exercise. The computational analysis of the VBM will be carried out using a finite element code (ABAQUS). The post-computational comparison of computed stress data for the keel bone to strength data of the keel bone will be conducted. We will either apply the maximum principal stress or the maximum principal strain failure criteria to compute values of a keel bone damage risk criterion as the ratio between computed principal stress (or strain) value and measured bone strength (or failure strain). We will execute the VBM for two types of conditions: (1) We will impose a range of loading conditions as these occur via palpation to represent external loading relevant to SA1. We thereby will apply external pressure to the VBM and support the VBM through contact surfaces representing the exam table; (2) We will investigate a loading condition on the keel bone as these occur during egg laying to represent internal loading relevant to SA2. We thereby will impose the load through the motion of an egg-shaped rigid body through the oviduct.

SA3d: VBM-Validation (PIs Siegmund, Nawrocki): We first validate the VBM on experiments on excised keel bone. These keel bones will be scanned in the CT- system, as well as conduct the RPI measurements of SA3b. Then, we will equip keel bones with sensors as described in SA1 and SA2 and place the keel bone specimens into a mechanical load frame. We will first apply a small magnitude of load for the bone to deform elastically only. We then compare measured sensor data

(contact pressure at the support and strain data) to respective data output computed from the VBM. An agreement of the two data sets of data within a 10% band of confidence is suggested as acceptable. Subsequently, the VBM will be extended to dynamic loading conditions using the explicit, time-marching algorithm of the ABAQUS FE-code. We thereby augment the VBM to include the tissue density and allow for dynamic loading conditions with initial velocity conditions and the collision of the VBM with rigid obstacles. The computed output of the VBM will be sampled at locations at which sensors are affixed to birds (SA1 and SA2). Computed time histories of acceleration and contact pressure data will be compared to corresponding measured data of external sensors (SA1). Computed time histories of strains on the keel bone will be compared to corresponding measured data of internal sensors (SA2). An agreement between computed and measured data will be used in the validation of the VBM and to obtain ranges of accuracy of the model.

Pitfalls and Limitations. The VBM as proposed seeks to focus on key features of the bird anatomy. It is, however, possible that the VBM would need to be endowed with information on internal organs, wings and legs. If our analysis indicates such, we will expand the anatomical reconstruction domain and the subsequent computational analysis. The VBM will employ a stress criterion for the assessment of KBD. If such a model is insufficient, we can expand the computational model to employ the extended finite element method whereby the formation of cracks in bone is explicitly accounted for.

Deliverables (for SA3).

- Validated virtual model for computations of keel bone stresses and strains
- Virtual model for slow (static) and fast (impact) loading on keel bone

SA4: EVALUATE KEEL BONE DAMAGE IN EXTENSIVE HOUSING CONSIDERING STOCKING DENSITY

SA 4 Outcomes: We will measure magnitudes of impact forces and bone strains in birds in extensive housing. We will use measured body acceleration data combined with video analysis of the bird behavior to extract keel bone damage critical events and assess keel bone damage risk through the application of the VBM. We will validate the keel bone risk assessment by CT imaging and in-vivo measurements of bone properties.

SA4a: Experimental Environment (PIs Karcher, Nawrocki): We will house egg laying hens in an enriched colony cage system under consideration of four levels of stocking density. We consider the application of our proposed approach in a study with stocking density as the housing design variable as an example of demonstration of the proposed approach to practice. In this environment, we enable birds to interact with an enriched environment as well as other conspecifics. Our ultimate objective is to house cohorts of 5, 10, 15, 20 birds at a time for observation periods of one week each. Our initial trial realization will employ five birds for 24 hours. This trial experiment will be repeated 5 times. We seek to repeat each of these weeklong experiments three times. Birds will be provided *ad libitum* access to water and feed. A video monitoring system will be set-up to continuously monitor the cage system during the daytime hours of the experiments. The video system consists of three video cameras, a DVR storage system and a video monitoring software. Birds will be subjected to CT scans prior to the one-week housing experiments as well as at the end points of the one week. Body weight data and egg production data will be collected for all birds on a daily basis. Image reconstruction as of SA3 will be employed to obtain the bird anatomy

and to assess any form of keel bone damage. All birds in each experiment will be equipped with acceleration sensors and the respective sensor data processing unit (backpack). Focal hens (1/5, 2/10, 3/15, 4/20) will be marked for identification and equipped with impact and keel bone sensors. Following SA1 these subgroups of birds will be outfitted with custom-made PVdF sensors to monitor forces due to impact. In addition, for the focal hens, strain sensors will be surgically attached to the keel bone using the procedure previously employed in SA2.

SA4b: Sensor Data and Analysis (PIs Karcher, Nawrocki, Siegmund): Acceleration data and video will be analyzed following a prior established protocol.¹³ Thereby, we identify critical acceleration events with behavior in the video analysis. We expect to again repeat the earlier finding of¹³ in that critical events are associated with collisions with perches in the cage, and we will specifically relate such events to stocking density. We then correlate impact force data with the acceleration data and the behavioral observation from video recordings. We hypothesize that the main acceleration events are similarly associated with the maxima in impact forces. Given Newton's second law of motion ($F = ma$) such a correlation is fundamentally relevant. We assume hens as deformable bodies. We then compute the corresponding skeletal deformation distance s as a new characteristic parameter, $s = \frac{1}{2} v^2 / F_{avg}$. The deformation distance describes the ability of a hen's body to absorb energy during the impact. We furthermore will measure bone strains during the critical impact events, but also during the egg laying process. We expect that impact related strains on the keel bone will cause tensile strains on the keel bone exterior surface, while the egg lay process would cause compressive strains on that surface. By comparison the impact related strains to egg laying related strain, we will then predict the likelihood of a keel bone fracture location. If impact dominates at a given stocking density, then cracking would occur from the keel bone outside while for cases where egg laying associated strains dominate, the defect would initiate from the inside. A validation of this hypothesis is possible by cross correlating CT images of damage to the measured strain ratio.

SA4c: Computational Analysis (PIs Karcher, Siegmund): Sensor data on acceleration in combination with the respective video will serve as relevant input to the analysis of impact events with the VBM. These VBM will be created primarily for the focal hens, based on individual CT scan images and the biomechanical properties as per SA3. We will create a virtual model of the enriched cage environment as rigid bodies such that a computational model then computes contact forces between the VBM and the rigid body. These contact forces are then compared to the measured impact forces to validate the computational model for the enhanced cage setting. The specific point measured strain sensor data will similarly be compared to the respective bone strain data to obtain further validation of the model. Once the VBM is validated, the complete multiaxial stress and strain state of the keel bone is available from the computational model. Such data possesses tensorial form in its raw computed state and is then postprocessed into the respective invariant based failure criteria (the maximum principal stress). The spatial and temporal distribution of the maximum principal stress then serves as the keel bone damage indicator. Computed keel bone damage indicator together with keel bone damaged assessed from CT images at post experiment states will be provided in a comparative analysis.

Pitfalls and Limitations. Sensors and VBM developed in SA1-3 may not be able to account for forces experienced by the birds in enriched colony cages. The more complex environment, increased bird number and interactions (bird-to-bird and bird-to-object) may impose unexpected complexities. If the housing environment is determined to be too noisy for high quality measurements, we would conduct the work of SA4 using conventional cage housed birds with

increased bird numbers per cage. Repeating the conventional cage environment will provide an ability to identify areas in the sensor or VBM modalities that can be adjusted for the next replication allowing for refinement of the process striving toward housing the birds in the enriched colony cage environment.

Deliverables (for SA4).

- Measurement of force experienced by the keel bone in a complex, realistic environment
- Correlation of force data with environment “events”, e.g. bird-to-bird and bird-to-object interactions
- Provide a demonstration example on housing design by considering stocking density as the design variable

Training and Integration

PI expertise in the proposed program covers distinctly different domains. This intellectual diversity provides unique opportunities for graduate student training with an emphasis on cross-cutting educational objectives. We will support three graduate students with their academic home in sensing and sensor development, animal sciences and mechanical engineering. The cross pollination of ideas from each of the disciplines will create unique professional profiles for each of the students involved in the research.

Communication of Data. These data will be presented at the Poultry Science, World’s Poultry Science annual meetings. We expect that this research will provide at least 5 research publications (*Poultry Science, PLoS ONE, Journal of Biomechanics, Computer Methods in Biomechanics and Biomedical Engineering, Frontiers in Animal Science*) and several scientific presentations.

Project Timetable (Will convert to Gantt chart for appeal)

	Year-1				Year-2				Year-3			
	1	2	3	4	1	2	3	4	1	2	3	4
SA1: External Measurements: Force	X	X	X	X	X	X	X	X				
SA2: Internal Measurements: Forces & Strains				X	X	X	X	X				
SA3: Virtual Bird Model					X	X	X	X	X	X	X	
SA4: Measurements of contributors to bone damage								X	X	X	X	X
Present data at conferences				X			X			X		X
Journal publications								X				X